INCH-POUND

O-S-571G
November 13, 1991
SUPERSEDING
O-S-571F
August 1, 1969

FEDERAL SPECIFICATION

SODIUM CARBONATE, ANHYDROUS, TECHNICAL

This specification is approved by the Commissioner of Federal Supply Service, General Services Administration, for use by all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers four types of technical grade anhydrous sodium carbonate (hereinafter referred to as sodium carbonate).

1.2 Classification. Sodium carbonate shall be of the following types as specified (see 6.2):

Type I	-	Ordinary or light ash
Type II	-	Medium ash
Type III	-	Dense or heavy ash
Type IV	-	Briquets

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Comments or suggestions pertaining to this specification should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423.

AMSC N/A

FSC 6810

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Federal Specifications:

UU-S-48	-	Sacks, Shipping, Paper
VV-L-800	-	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature)
PPP-B-35	-	Bags, Textile, Shipping, Burlap, Cotton and Waterproof Laminated
PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-D-705	-	Drum, Shipping and Storage: Steel, 16 and 30 Gal. Capacity
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-F-320	-	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallon)
PPP-P-1660	-	Pallet, Expendable

Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies)
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA, Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

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Military Specifications:

MIL-B-117	-	Bags, Sleeves and Tubing
MIL-P-15011	-	Pallets, Material Handling, Wood Post Construction, 4-Way Entry

Military Standards:

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTM Standards:

D 501	-	Sampling and Chemical Analysis of Alkaline Detergents
D 3950	-	Strapping, Nonmetallic (and Joining Methods)
D 3953	-	Strapping, Flat Steel and Seals

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

"National Motor Freight Classification"

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent

"Uniform Freight Classification"

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Chemical and physical characteristics.** Sodium carbonate shall conform to the applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. **Chemical and physical characteristics**

Characteristic	Type I	Type II	Type III	Type IV	Test paragraph
Loss on heating, maximum percent by weight	1.0	1.0	1.0	1.0	4.2.4.1
Water-insoluble matter in dried material, maximum percent by weight	0.25	0.25	0.25	0.25	4.2.4.2
Total alkalinity in dried material, minimum percent by weight	99.2	99.2	99.2	98.0	4.2.4.3
Apparent density, pounds per cubic foot	29 to 37	42 to 62	54 to 75	------	4.2.4.4

3.1.1 **Weight of briquets (type IV).** Twenty-five briquets shall weigh 16 ± 1 ounces (oz) when tested as specified in 4.2.4.5.

3.2 **Material Safety Data Sheets.** Material Safety Data Sheets for sodium carbonate shall be prepared and submitted by the contractor in accordance with FED-STD-313 (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.4.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium carbonate of one type produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.5).

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with table II. The sample unit shall be one filled unit pack or packing container, as applicable, ready for shipment.

TABLE II. **Sampling for packaging examination**

Number of containers in batch or lot	Number of sample containers
1 or 2	all
3 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22
over 10,000	29

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table III. A representative specimen of approximately 1 pound (lb) of type I, II, or III sodium carbonate or approximately 2 lb of type IV sodium carbonate shall be removed from each sample container and placed in a suitable clean, dry closed container labeled to identify the lot and container from which it was taken.

TABLE III. **Sampling for sodium carbonate test**

Number of containers in batch or lot	Number of sample containers
1 or 2	all
3 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
Over 7,000	10

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4.2.3 **Inspection procedure.**

4.2.3.1 **For examination of packaging.** Sample containers shall be examined for the characteristics listed below. Failure of any sample container to conform to all characteristics shall be cause for rejection of the lot represented.

(a) Contents per container
(b) Container
(c) Container closure
(d) Container free of damage and leaks
(e) Interior bag
(f) Interior bag closure
(g) Wood preservation evident and correct (type IV, level A only)
(h) Marking evident, correct and legible

4.2.3.2 **For test of types I, II, and III sodium carbonate.** Approximately equal portions of the specimens taken in 4.2.2.2, shall be thoroughly mixed to form a composited specimen of no less than 1 lb. The composited specimen shall be tested as specified in the applicable portions of 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 **For test of type IV sodium carbonate.** Twenty-five briquets of each specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4.5. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test by any specimen shall be cause for rejection of the lot represented. The remaining specimen material shall then be composited and tested as specified in 4.2.3.2.

4.2.4 **Tests.** Tests shall be conducted as follows:

4.2.4.1 **Loss on heating.** Determine percent loss on heating in accordance with ASTM D 501, sections 18 and 19.

4.2.4.2 **Water-insoluble matter.** Dry approximately 25 grams (g) of the specimen for 1 hour at 150° to 155°C and then determine percent water-insoluble matter in accordance with ASTM D 501, sections 29 and 30.

4.2.4.3 **Total alkalinity.** Dry approximately 2 g of the specimen for 1 hour at 150° to 155°C and then determine percent alkalinity in accordance with ASTM D 501, sections 20, 21, and 22.

4.2.4.4 **Apparent density.** Determine apparent density in accordance with ASTM D 501, sections 31 and 32.

4.2.4.5 **Weight of briquets (type IV).** Weigh 25 briquets collectively to the nearest 0.1 oz.

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5. PACKAGING

5.1 Unit packing. Unit packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. Five pounds (+1/2 or -0 ounces) of type I, II, or III sodium carbonate shall be unit packed in a paperboard box conforming to variety 2, style I, type A, class B, single wax dipped of PPP-B-566. The box shall be closed as specified in the appendix to PPP-B-566.

5.1.2 Level C. Five pounds (+1/2 or -0 ounces) of type I, II, or III sodium carbonate shall be unit packed in a paperboard box conforming to variety 1, style I, type optional, class B of PPP-B-566. The box shall be closed as specified in the appendix to PPP-B-566.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Five-pound quantity. Twelve boxes of sodium carbonate, unit packed as specified in 5.1.1, shall be packed in either a nailed wood box conforming to class 2, style 4 or 4-1/2, for a type 2 load of PPP-B-621; a cleated-plywood box conforming to type overseas, style I of J, for a type 2 load of PPP-B-621; or a wirebound box conforming to class 3, style optional, for a type 2 load of PPP-B-585. The box shall be closed and reinforced in accordance with the requirements in the appendix to the applicable box specification.

5.2.1.2 Twenty-five-pound quantity. Twenty-five pounds (+1/4 or -0 pounds) of type I, II, or III sodium carbonate shall be packed in a metal pail conforming to type II, class 4 of PPP-P-704 with a heat sealable, interior packaging bag conforming to type I, class B, style 2 of MIL-B-117. The interior surfaces of the pail shall be fog sprayed with a rust-preventative oil conforming to VV-L-800 which shall be applied in compliance with the requirements specified in PPP-D-705. Closure of the interior bag shall be accomplished by heat sealing or by twisting the open end of the bag and tying it in a knot. Closure of the pail shall be in compliance with the appendix to PPP-P-704.

5.2.1.3 One hundred-pound quantity. One hundred pounds (+1/2 or -0 pounds) of type I, II, or III sodium carbonate shall be packed in a 25-gallon, fully removable head metal drum conforming to Department of Transportation (DOT) specification 17H, with a heat-sealable interior packaging bag conforming to type I, class B, style 2 of MIL-B-117. The interior surfaces of the drum shall be treated as specified in 5.2.1.2. Exterior surfaces of the drum, including the cover and bottom head, shall be coated and painted as specified for the drums of PPP-D-729. Closure of the interior bag shall be as specified in 5.2.1.2. Closure of the drum shall be in compliance with DOT requirements.

5.2.1.4 Three hundred-pound quantity. Three hundred pounds (+1 or -0 pounds) of type I, II, or III sodium carbonate shall be packed in a 55-gallon, fully removable head metal drum conforming to type IV of PPP-D-729 with a heat-sealable interior packaging bag conforming

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to type I, class B, style 2 of MIL-B-117. The interior surfaces of the drum shall be fog sprayed with a rust-preventative oil as specified in PPP-D-729. Closure of the bag shall be as specified in 5.2.1.2. Closure of the drum shall be in compliance with PPP-D-729.

5.2.1.5 Briquets. Fifty pounds (+1/2 or -0 pounds) of type IV sodium carbonate shall be packed in a heat-sealable bag conforming to type I, class B, style 2 of MIL-B-117. The filled and sealed bag shall be packed in a cleated-plywood box conforming to type overseas, style I or J of PPP-B-601 having a fiberboard liner constructed of material conforming to type CF, grade W5c of PPP-F-320. The liner shall completely separate the bagged contents from the surfaces of the box. Closure and reinforcement of the box shall be in compliance with requirements in the appendix to PPP-B-601. Unless otherwise specified, all unfinished, unpainted wood parts, including cleats, shall be treated with water-repellent wood preservative as specified for the plywood panels in PPP-B-601.

5.2.2 Level B.

5.2.2.1 One hundred-pound quantity. One hundred pounds (+1/2 or -0 pounds) of sodium carbonate shall be packed in a bag conforming to PPP-B-35, type III or in a sack conforming to UU-S-48 for soda ash.

5.2.3 Level C. Quantities of sodium carbonate, as specified (see 6.2), of the type specified (see 6.2), shall be packed in containers which comply with the requirements of the Uniform Freight Classification or National Motor Freight Classification rating, rules, and regulations, as applicable, or other carrier rules as may be applicable to the mode of transportation. Containers shall be capable of being stacked and of supporting superimposed loads during shipment and shall assure sage delivery to destination without damage to contents.

5.3 Marking. In addition to the marking specified in 5.3.1 and any marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129 or Fed. Std. No. 123, as applicable.

5.3.1 Precautionary marking. All containers shall be durably and legibly marked as follows:

AVOID INHALATION OF DUST
AVOID CONTACT WITH SKIN AND EYES

5.4 Palletization. When specified in the contract or order, shipping containers shall be palletized as specified below. Unless otherwise specified, pallets and associated wood storage aids shall be treated with wood preservative as specified in MIL-P-15011 (see 6.2).

5.4.1 Twenty-five-pound quantity. The 25-pound containers of sodium carbonate shall be palletized in three rows of four containers each, in four tiers, on a style 1, type I, class 1 pallet of MIL-P-15011 with top and bottom caps. Strapping shall conform to ASTM D 3953,

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zinc-coated, and shall consist of three equally spaced, 1.25-inch by 0.035-inch galvanized straps running across the top of the top cap and under the deck boards of the pallet in the 40-inch direction, and horizontal strapping running completely around each tier of containers as specified for load type III in MIL-STD-147.

5.4.2 One hundred-pound quantity.

5.4.2.1 Drums. When specified (see 6.2), 25-gallon drums shall be palletized on a style 1, type I, class 1 pallet of MIL-P-15011 in accordance with the requirements for load type XII. Drums shall be positioned so as to attain maximum utilization of the 40-inch by 48-inch pallet and shall not exceed a height of 54 inches. Unitizing strapping shall conform to ASTM D 3953, zinc-coated, 1.25-inch by 0.035-inch.

5.4.2.2 Sacks. When specified (see 6.2), sacks of sodium carbonate shall be packed on a pallet conforming to PPP-P-1660 type II, style optional, class B or C, grade 2, any size except h. The load shall be secured to the pallet with flat steel or non-metallic strapping conforming to ASTM D 3950 or ASTM D 3953. When steel strapping is used, metal plastic or fiberboard edge protectors shall be placed under the strap at the edge point. A minimum of two straps shall be placed around the top, bottom and sides of the load and two straps around the top, bottom and ends of the load. Steel strapping applied to a load at right angles to each other and in the same plane shall be prevented from lateral movement and displacement by the application of steel crossties of a size appropriate for the flat steel strapping used. Alternatively, the load shall be secured to the pallet by the shrink film process. Thermoplastic films of 6 mil polyethylene, 5.5 mil reinforced polyethylene, or 2.5 mil polyvinyl chloride shall be used. Shrinking of the film shall be in accordance with industry practice.

5.4.3 Briquets. Boxes containing type IV sodium carbonate shall be palletized on a style 1, type I, class 1 pallet of MIL-P-15011 in accordance with the container pattern in the appendix to MIL-STD-147 most effective for attaining maximum utilization of the 40-inch by 48-inch pallet. One horizontal strap of 0.75-inch by 0.023-inch galvanized steel, conforming to ASTM D 3953, zinc-coated, shall be applied to each tier and three primary tie-down straps. Galvanized straps of 1.25-inch width and 0.035-inch thickness conforming to ASTM D 3953, zinc-coated, shall be equally spaced and shall run across the top of the load and under the deck board of the pallet in 40-inch direction.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sodium carbonate covered by this specification is intended for various washing, cleaning, and scouring processes, with or without soap, as conditions may require when a moderately strong alkaline material is desired, and for the treatment of water. This specification is not intended to apply to soda ash used in glass making.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Type of sodium carbonate required (see 1.2)
(c) Level of unit packing and packing required (see 5.1 and 5.2)
(d) Unit quantity required (see 5.1 and 5.2)
(e) Additional marking, if required (see 5.3)
(f) Palletization and preservation thereof, if required (see 5.4).

6.3 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent mailing addresses for submissions of data are listed in FED-STD-313.

6.4 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.6 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Subject term (key word) listing.

alkaline
cleaning processes
scouring processes

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MILITARY INTERESTS:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD, ME, AR
DLA - GS

User activities:

Army - SM
Navy - MC, SH, YD

CIVIL AGENCY COORDINATING ACTIVITIES:

COM-NIST
DOI-MIN
DOT-ACO
DOT-FAA
GSA-FSS (9FTE-10)
NASA-JFK

Preparing activity:

Army - EA

Project No. 6810-1239

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